

Numbers ke Prakar

Types of Numbers - Aasan Hinglish Notes, Examples aur Practice Q's

Maths me numbers ko alag-alag families me baanta jata hai, aur har family pichli wali ke upar bani hoti hai. Is guide me har type ko simple Hinglish me examples ke saath samjhaya gaya hai, fir solved examples aur end me 22 practice questions ke step-by-step solutions diye gaye hain.

EXAM * aise green tag wale topics exams me baar-baar aate hain

1. Natural Numbers (N)

Definition: Ginti (counting) wale numbers jo 1 se shuru hote hain aur aage badhte rehte hain: 1, 2, 3, 4, ...

Symbol: N

Example: 1, 2, 3, 4, 5, ... (jaise class me students ki sankhya)

Yaad rakho:

- Sabse chhota natural number 1 hai.
- Zero (0) ek natural number NAHI hai.
- Ye hamesha positive whole numbers hote hain.

2. Whole Numbers (W)

Definition: Saare natural numbers + 0 milakar: 0, 1, 2, 3, 4, ...

Symbol: W

Example: 0, 1, 2, 3, 4, ...

Yaad rakho:

- Whole numbers = Natural numbers + 0.
- Sabse chhota whole number 0 hai.
- Har natural number ek whole number hai (par 0 whole hai, natural nahi).

3. Integers (Z)

EXAM * Exam me CONSTANTLY aata hai

Definition: Saare whole numbers + unke negatives: ..., -3, -2, -1, 0, 1, 2, 3, ...

Symbol: Z

Example: -100, -5, -1, 0, 3, 47

Yaad rakho:

- Isme positive, negative aur zero teeno aate hain.
- Integers me koi fraction ya decimal nahi hota.
- Zero ek integer hai jo na positive hai na negative.

4. Rational Numbers (Q)

EXAM * Exam me CONSTANTLY aata hai

Definition: Wo numbers jinhe p/q fraction ke roop me likha ja sake, jahan p aur q integers hon aur q zero ke barabar na ho.

Symbol: Q

Example: $1/2$, $-3/4$, 5 ($= 5/1$), 0.75, 0.333... ($= 1/3$)

Yaad rakho:

- Jo decimals khatam ho jayein (terminating) ya repeat hon, wo rational hain.
- Har integer bhi ek rational number hai.
- Denominator q kabhi 0 nahi ho sakta.

5. Irrational Numbers

EXAM * Exam me CONSTANTLY aata hai

Definition Wo numbers jinhe simple fraction p/q me NAHI likha ja sakta. Inka decimal part kabhi khatam nahi hota aur repeat bhi nahi karta.

Example: $\pi = 3.14159\dots$, $\sqrt{2} = 1.41421\dots$, $e = 2.71828\dots$

Yaad rakho:

- Ye non-terminating aur non-repeating decimals hote hain.
- Non-perfect squares ke square root irrational hote hain ($\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$).
- π aur e sabse famous irrational numbers hain.

6. Real Numbers (R)

EXAM * Exam me CONSTANTLY aata hai

Definition Saare rational aur irrational numbers milakar. Jo bhi number number-line par rakha ja sake, wo real number hai.

Symbol: $\mathbb{R}$

Example: -2, 0, $1/2$, 7.5, $\sqrt{2}$, π

Yaad rakho:

- Real numbers = Rational numbers + Irrational numbers.
- Roz-marra ke lagbhag saare numbers real numbers hote hain.
- $\sqrt{-1}$ jaise numbers real NAHI hote - unhe imaginary numbers kehte hain.

7. Even aur Odd Numbers

Definition Even numbers 2 se पूरी तरह divide ho jate hain; odd numbers 2 se divide nahi hote.

Example: Even: 2, 4, 6, 8, 10 | Odd: 1, 3, 5, 7, 9

Yaad rakho:

- Even numbers ke end me 0, 2, 4, 6 ya 8 aata hai.
- Odd numbers ke end me 1, 3, 5, 7 ya 9 aata hai.
- Zero (0) ek even number hai.

8. Prime aur Composite Numbers

EXAM * Exam me CONSTANTLY aata hai

Definition Prime number ke exactly do factors hote hain (1 aur khud wo number). Composite number ke do se zyada factors hote hain.

Example: Prime: 2, 3, 5, 7, 11 | Composite: 4, 6, 8, 9, 10

Yaad rakho:

- 2 hi ekmatra even prime number hai.
- 1 na prime hai na composite.
- Sabse chhota prime 2 hai; sabse chhota composite 4 hai.

9. Aur Zaroori Types (Exam Favourite)

EXAM * Exam me CONSTANTLY aata hai

Co-prime Numbers Do numbers jinka HCF sirf 1 ho. Jaroori nahi ki wo khud prime hon. Jaise (8, 9), (15, 16).

Consecutive Numbers Lagatar aane wale numbers jaise 5, 6, 7. Do consecutive numbers ka antar hamesha 1 hota hai.

Twin Primes Aise do prime numbers jinke beech sirf 2 ka antar ho. Jaise (3, 5), (11, 13), (17, 19).

Perfect Numbers Wo number jiske apne factors ka jod (khud ko chhodkar) usi number ke barabar ho. Jaise $6 = 1 + 2 + 3$.

Number Line Ek seedhi line jisme left me negative, beech me 0, aur right me positive numbers hote hain. Har real number is par ek point hota hai.

Quick Revision Summary

Number families ek dusre ke andar aise fit hoti hain:

Natural < Whole < Integers < Rational < Real

Irrational numbers bhi Real hain, par Rational se alag.

Real Numbers = Rational + Irrational (number line ke saare numbers).

Solved Examples - Samjho Kaise Pehchante Hain

Neeche 10 examples step-by-step solve karke dikhaye gaye hain, taaki number classify karna aur reasoning clear ho jaye.

Example 1: -7 kis type ka number hai?

- Step: Negative hai, isliye natural ya whole nahi.
- Step: Bina fraction ke poora number hai -> Integer.
- Step: $-7 = -7/1$ likh sakte hain -> Rational bhi hai.

Answer: Integer (aur Rational)

Example 2: $\sqrt{16}$ rational hai ya irrational?

- Step: Pehle value nikalo: $\sqrt{16} = 4$.
- Step: 4 ek whole number hai -> $4/1$ (p/q form).
- Step: (Trick: 16 perfect square hai, isliye root rational nikla!)

Answer: Rational

Example 3: 0.1010010001... rational ya irrational?

- Step: Decimal kabhi khatam nahi hota aur koi pattern repeat nahi karta.

Answer: Irrational

Example 4: 0.6 ko p/q form me likho.

- Step: $0.6 = 6/10$.
- Step: HCF 2 se simplify: $6/10 = 3/5$.

Answer: $3/5$

Example 5: Kya 14 aur 15 co-prime hain?

- Step: $14 = 2 \times 7$, $15 = 3 \times 5$.
- Step: Koi common factor nahi -> HCF = 1.

Answer:Haan, co-prime hain

Example 6: 1 prime hai, composite hai, ya dono nahi?

- Step: Prime ke 2 factors hote hain; composite ke 2 se zyada.
- Step: 1 ka sirf ek factor hai (khud).

Answer:Dono nahi (neither)

Example 7: $\frac{1}{3}$ aur $\frac{1}{2}$ ke beech ek rational number batao.

- Step: Dono ka average lo: $(\frac{1}{3} + \frac{1}{2}) / 2$.
- Step: $= (\frac{5}{6}) / 2 = \frac{5}{12}$ (ye $\frac{1}{3}$ aur $\frac{1}{2}$ ke beech hai).

Answer: $\frac{5}{12}$

Example 8: Sabse chhota 2-digit prime number?

- Step: 10 composite hai (2×5).
- Step: 11 ke factors sirf 1 aur 11.

Answer:11

Example 9: Kya 2 ke alawa koi even prime hai?

- Step: Har dusra even number 2 se divide hota hai.
- Step: Isliye uske 2 se zyada factors $\rightarrow$ composite.

Answer:Nahi, sirf 2

Example 10: pi ko classify karo.

- Step: $\pi = 3.14159\dots$ non-terminating, non-repeating.
- Step: Isliye Irrational; aur number line par hai to Real bhi.

Answer:Irrational (aur Real)

Practice Questions

Pehle khud solve karne ki koshish karo. Neeche har question ka step-by-step solution diya gaya hai.

Q1. Sabse chhota natural number kaunsa hai?

Q2. Sabse chhota whole number kaunsa hai?

Q3. Kya 0 ek natural number hai? (Haan/Nahi)

Q4. Kya 0 ek whole number hai? (Haan/Nahi)

Q5. -5 kis type ka number hai (natural / whole / integer)?

Q6. Kya har natural number ek whole number hota hai?

Q7. $\frac{3}{4}$ kis type ka number hai?

Q8. $\sqrt{2}$ rational hai ya irrational?

Q9. π (3.14159...) rational hai ya irrational?

Q10. 7 ko $\frac{p}{q}$ form me likho.

Q11. Kya 0.75 rational hai? (Haan/Nahi)

Q12. Sabse chhota prime number kaunsa hai?

Q13. Kya 1 prime hai ya composite?

Q14. 2 ke alawa koi aur even prime number hai? (Haan/Nahi)

Q15. 4, 6 aur 8 - ye prime hain ya composite?

Q16. Kya 0 even hai ya odd?

Q17. 17 prime hai ya composite?

- Q18.** Real numbers kis-kis se milkar bante hain?
- Q19.** Kya $\sqrt{-1}$ ek real number hai?
- Q20.** -3, 0 aur 5 - ye sab kis ek hi category me aate hain?
- Q21.** 9 ke saare factors likho. Kya 9 composite hai?
- Q22.** 0.333... (repeating) rational hai ya irrational?

Step-by-Step Solutions

Q1. Sabse chhota natural number?

- Step: Natural numbers 1 se shuru hote hain.

Answer:1

Q2. Sabse chhota whole number?

- Step: Whole numbers 0 se shuru hote hain.

Answer:0

Q3. Kya 0 ek natural number hai?

- Step: Natural 1 se shuru; 0 unme nahi aata.

Answer:Nahi

Q4. Kya 0 ek whole number hai?

- Step: Whole = Natural + 0, isliye 0 whole hai.

Answer:Haan

Q5. -5 kis type ka number hai?

- Step: Negative hai -> natural/whole nahi.
- Step: Bina fraction ke poora -> Integer.

Answer:Integer (aur Rational)

Q6. Kya har natural number whole hota hai?

- Step: Whole = Natural + 0, to har natural whole hai.

Answer:Haan

Q7. $\frac{3}{4}$ kis type ka number hai?

- Step: $\frac{p}{q}$ form me hai ($q \neq 0$).

Answer:Rational

Q8. $\sqrt{2}$ rational ya irrational?

- Step: $\sqrt{2} = 1.414...$ non-terminating, non-repeating.

Answer:Irrational

Q9. pi rational ya irrational?

- Step: $\pi = 3.14159...$ kabhi khatam/repeat nahi hota.

Answer:Irrational

Q10. 7 ko $\frac{p}{q}$ form me likho.

- Step: Kisi whole number ko $\frac{\quad}{1}$ likh sakte hain.

Answer: $\frac{7}{1}$

Q11. Kya 0.75 rational hai?

- Step: $0.75 = \frac{75}{100} = \frac{3}{4}$ (terminating decimal).

Answer:Haan

Q12. Sabse chhota prime number?

- Step: 1 prime nahi; 2 ke sirf 2 factors (1, 2).

Answer:2

Q13. 1 prime hai ya composite?

- Step: 1 ka sirf ek factor hai (khud).

Answer:Dono nahi (neither)

Q14. 2 ke alawa koi even prime?

- Step: Baaki even numbers 2 se divisible $\rightarrow$ 2 se zyada factors.

Answer:Nahi

Q15. 4, 6, 8 prime ya composite?

- Step: Sabke 2 se zyada factors hain.

Answer:Composite

Q16. 0 even hai ya odd?

- Step: 0 ko 2 poora divide karta hai ($0/2 = 0$).

Answer:Even

Q17. 17 prime ya composite?

- Step: 17 ke factors sirf 1 aur 17.

Answer:Prime

Q18. Real numbers kis-kis se bante hain?

- Step: Real = Rational + Irrational.

Answer:Rational + Irrational

Q19. Kya $\sqrt{-1}$ real number hai?

- Step: Kisi real number ka square negative nahi hota.

Answer:Nahi (imaginary)

Q20. -3, 0, 5 kis category me aate hain?

- Step: Teeno bina fraction ke poore numbers hain.

Answer:Integers

Q21. 9 ke factors; kya 9 composite hai?

- Step: Factors: 1, 3, 9 $\rightarrow$ 2 se zyada factors.

Answer:1,3,9; Haan composite

Q22. 0.333... rational ya irrational?

- Step: Repeating decimal = $1/3$, p/q form me likha ja sakta hai.

Answer:Rational